MGM Macau and Cotai

Resource Efficiency Management Report

July 2022

Energenz Consulting Limited

EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to July 2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-5.9%**	**-5.9%**	-	-	**-**
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.8%**	**-10.0%**	-	-	**-**

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2022.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to July 2022
Macau	2019	15%	**-8.6%**
Cotai	2019	15%	**-16.5%**

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

Highlighted Changes in Operation

Macau

1. Due to the humid weather from the beginning of May 2022, usage of heat pump system for L2M boiler for the floor below L19 have been increased for the sake of dehumidification.
2. As part of the energy saving strategy during COVID low occupancy periods, MGM Macau has implemented the followings in July 2022:
    a. Casino AHUs are adjusted for reducing the ACH and hot fresh air intake.
    b. Suspended the unoccupied smoking lounge at L2 VIP gaming (SunCity, Guangdong etc.)
    c. Increased space temperature and reduced supply air flow for suspended restaurants.
    d. Increased space temperature and reduced supply air flow for suspended kitchens. Kitchen exhaust fan, UV light and the EP has been switched off.
    e. Increased space temperature and reduced supply air flow for Grand Praca.
    f. Hotel tower empty guest rooms lighting are switched off, FCU running at low speed and increased space temperature.
    g. Minimized the lighting and ventilation in car park areas.
    h. Switched off the hotel fountain jet pump at day time and underwater & statue lighting at night time, swimming pool and jacuzzi.
    i. Increased the Chiller CHWST setpoint by 0.5°C
    j. Decreased the hot water temperature by 2°C

Cotai

1. MGM has adjusted the HRU-1F-1,2,3 fan speed from 35 Hz to 25 Hz and reduced the fresh air in-take to AHU-GF-05,06,07. This could reduce the heating load to the cooling coil of the AHUs.
2. Due to the COVID low occupancy periods, the below practices have been proceeded in July 2022:
    a. Casino AHUs are adjusted for reducing the ACH and hot fresh air intake.
    b. Suspended the unoccupied smoking lounge at 1F junket gaming (SunCity, Guangdong etc.)
        i. All exhaust fans command is set to OFF.
        ii. All door status alarm & diff. pressure alarm alert emails are isolated/disabled, and the point signals command is set to NORMAL
        iii. All alarm buzzer commands are set to OFF.
        iv. The point signal of diff. pressure value command is set to < -12.5Pa
    c. Increased space temperature and reduced supply air flow for suspended restaurants.
    d. Increased space temperature and reduced supply air flow for suspended kitchens. Kitchen exhaust fan, UV light and the EP has been switched off.

e. Increased space temperature and reduced supply air flow for spectacle.
f. Hotel towers empty guest rooms lighting are switched off, FCU running at low speed and increased space temperature.
g. Minimized the lighting and ventilation in car park areas.
h. Switched off both hotel towers and mansion façade lighting
i. Switched off the landscape water feature, swimming pool and jacuzzi.
j. Increased the Chiller CHWST setpoint by 0.5°C
k. Decreased the hot water temperature by 2°C

Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m^2)	% Change Compared to 2021
Macau	**24.4**	**-3.0%**
Cotai	**23.8**	**-9.4%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m^3/ m^2)	% Change Compared to 2021
Macau	**0.19**	**-10.5%**
Cotai	**0.12**	**-24.1%**

Note: Cotai Area: 374,138 m^2 Macau Area: 211,545 m^2

ISO 50001 Certification Audit

Macau

The HKQAA 2022 surveillance audit was completed on 30 March 2022. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

Cotai

The HKQAA 2022 recertification audit will be scheduled on 29 & 30 September 2022.

Target Milestones for the Year 2022

The following activities are planned for completion this year:

- To perform ISO 50001 2022 Internal and External Audit for MGM Macau and MGM Cotai
- To deploy and fine-tune the DGLux dashboard with heatmap development extension
- To perform 2021 Carbon and Waste Audit for MGM Macau and MGM Cotai
- To assist MGM with the COTAI phase 2 design and planning
- To investigate and implement the below list of projects during the Year 2022

Table 5. MGM Macau Year 2022 Projects

Item	Initiatives
Electricity & LPG	
1	Chilled Water Pump Retrofit (Phase 1)
2	AHU Static Pressure Set-Point Reset (Phase 2)
3	PICCV Implementation (Phase 3)
4	Air Side Equipment EC Fan/Motor Replacement
5	VAV Space Temperature Setpoint Optimization with Heatmap Control
6	Heatmap Phasing Implementation (Stage 3, 9a, 9b, 10a)
7	Neon Light LED Replacement (Phase 2)
8	Replacement of L2M Steam-to-Water Heat Exchanger to Fully Utilize Heat Pump
Water	
9	CT Conductivity Blowdown Optimization

Table 6. MGM Cotai Year 2022 Projects

Item	Initiatives
Electricity & NG	
1	Chiller Plant Control Optimization (CHWP/CDWP/CT)
2	Hot Water System Upgrade 1. Increase Max. No of Plate for Pre-heat Compact Unit 2. Upgrade Booster Compact Unit 3. Install Electric Heater for Booster Calorifier
3	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)
4	PICCV Implementation (Phase 3)
5	Hotel Towers Building Pressurization and EAF Optimization
6	Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization
7	Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2)
8	VAV Space Temperature Setpoint Optimization with Heatmap Control
9	Heatmap Phasing Implementation (Stage 5, 6, 7, 8a, 8b, 10b)
10	Centralized RO Pump Set Replacement (10 Units)
Water	
11	Extension of CDP Water Recycling
12	Rainwater Harvesting for CT Make Up (Phase 1)
13	Non-potable Centralized RO Optimization for Cooling Tower

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ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, and China 3 Star Green Building Operation Label.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1. Portfolio Performance Summary

1.1 Electricity Use Intensity (EUI)

Macau's electricity use intensity (EUI) in July 2022 had decreased by **15.8%** compared to the same month in 2021 and decreased by **3.0%** in 2022 compared to 2021. Meanwhile, Cotai's EUI in July 2022 has decreased by **23.4%** compared to the same month in 2021 and decreased by **9.4%** in 2022 compared to 2021.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2021 Actual EUI (kWh/m²)	Year 2022 Actual EUI (kWh/m²)	% Change	Year 2021 Actual EUI (kWh/m²)	Year 2022 Actual EUI (kWh/m²)	% Change
January	20.4	22.6	+11.0%	21.8	22.3	+2.1%
February	19.8	20.2	+1.8%	20.4	20.1	-1.4%
March	23.3	24.4	+4.5%	24.6	23.6	-4.0%
April	23.8	24.3	+1.8%	25.4	24.0	-5.6%
May	28.9	25.9	-10.5%	30.6	25.6	-16.3%
June	28.7	27.3	-4.9%	29.7	27.0	-9.3%
July	30.8	25.9	-15.8%	31.7	24.3	-23.4%
August						
September						
October						
November						
December						
Average	**25.1**	**24.4**	**-3.0%**	**26.3**	**23.8**	**-9.4%**

1.2 Water Use Intensity (WUI)

Macau's water use intensity (WUI) in July 2022 decreased by **32.6%** compared to the same month in 2021 and decreased by **10.5%** in 2022 compared to 2021. Meanwhile, Cotai's WUI in July 2022 decreased by **46.2%** compared to the same month in 2021 and decreased by **24.1%** in 2022 compared to 2021.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2021 Actual WUI (m^3/m^2)	Year 2022 Actual WUI (m^3/m^2)	% Change	Year 2021 Actual WUI (m^3/m^2)	Year 2022 Actual WUI (m^3/m^2)	% Change
January	0.19	0.18	-7.2%	0.116	0.119	+3.2%
February	0.19	0.20	+1.1%	0.14	0.12	-10.9%
March	0.20	0.19	-7.4%	0.15	0.12	-19.3%
April	0.24	0.21	-9.1%	0.18	0.14	-21.2%
May	0.25	0.23	-7.7%	0.21	0.15	-28.2%
June	0.21	0.19	-9.3%	0.18	0.12	-30.7%
July	0.22	0.15	-32.6%	0.19	0.10	-46.2%
August						
September						
October						
November						
December						
Average	**0.21**	**0.19**	**-10.5%**	**0.16**	**0.12**	**-24.1%**

2. MGM Macau Performance

2.1 Electricity Consumption

The actual electricity consumption and cost in July 2022 decreased by **15.8%** and **12.7%** respectively compared to the same month in 2021. To date, MGM Macau has decreased the actual electricity consumption by **3.0%** and associated cost increased by **1.1%** respectively.

Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	4,310,160	4,784,280	+11.0%	4,717,260	5,519,350	+17.0%	+15.8%
February	4,190,760	4,266,240	+1.8%	4,591,170	4,942,590	+7.7%	-90.8%
March	4,931,880	5,152,320	+4.5%	5,387,480	5,938,160	+10.2%	-6.9%
April	5,042,160	5,131,680	+1.8%	5,808,340	6,183,420	+6.5%	-11.4%
May	6,122,640	5,478,840	-10.5%	7,034,530	6,595,030	-6.2%	-32.6%
June	6,072,960	5,777,040	-4.9%	7,681,070	7,567,480	-1.5%	-1.0%
July	6,506,760	5,480,880	-15.8%	8,217,570	7,172,780	-12.7%	+4.1%
August							
September							
October							
November							
December							
Total/Average	**37,177,320**	**36,071,280**	**-3.0%**	**43,437,420**	**43,918,810**	**+1.1%**	**-13.1%**

Important note: *Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Due to the adjusted operations during COVID low occupancy, the overall site electricity has decreased significantly when compared with the same month last year.

Table 10 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in July 2022 increased by **3.6%** compared to the same month in 2021.

Table 10. Macau - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2021 – Electricity Rate (MOP/kWh)	Year 2022 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.09	1.15	+5.4%
February	1.10	1.16	+5.7%
March	1.09	1.15	+5.5%
April	1.15	1.20	+4.6%
May	1.15	1.20	+4.8%
June	1.26	1.31	+3.6%
July	1.26	1.31	+3.6%
August			
September			
October			
November			
December			
Total/Average	**1.16**	**1.21**	**+4.7%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in July 2022 decreased by **20.1%** compared to 2019 normalized consumption in the same month.

Table 11. Macau - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2022 Actual Electricity Consumption (kWh)	% Change - Normalized
January	4,790,450	4,784,280	-0.1%
February	4,631,715	4,266,240	-7.9%
March	5,342,712	5,152,320	-3.6%
April	5,631,086	5,131,680	-8.9%
May	5,930,696	5,478,840	-7.6%
June	6,468,840	5,777,040	-10.7%
July	6,859,773	5,480,880	-20.1%
August			
September			
October			
November			
December			
Total/Average	**39,655,272**	**36,071,280**	**-9.0%**

2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption and cost in July 2022 decreased by **37.0%** and **6.9%** respectively compared to the same month in 2021. To date, MGM Macau has increased the actual LPG consumption by **15.4%** and associated costs increased by **51.8%**.

Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	121,398	121,221	-0.1%	947,708	1,327,490	+40.1%	+56.0%
February	74,797	146,623	+96.0%	632,778	1,450,103	+129.2%	-52.6%
March	84,131	91,040	+8.2%	711,751	900,382	+26.5%	-5.4%
April	77,702	85,743	+10.3%	652,357	858,756	+31.6%	-28.6%
May	71,294	82,704	+16.0%	578,906	992,447	+71.4%	-26.8%
June	62,104	77,044	+24.1%	504,285	924,531	+83.3%	-1.2%
July	71,398	44,991	-37.0%	579,753	539,887	-6.9%	-0.8%
August							
September							
October							
November							
December							
Total/Average	**562,824**	**649,365**	**+15.4%**	**4,607,539**	**6,993,595**	**+51.8%**	**-12.8%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption decrease below:

- Boilers LPG consumption decreased by 32.8% in July 2022 in July 2022 compared with the same month in 2021.
- Kitchen LPG Consumption has decreased by 70.3% in July 2022 compared with the same month in 2021 due to COVID low occupancy during this period.

Table 13 presents the comparison of LPG rates in the years 2021 and 2022. The LPG rate in July 2022 increased by **47.8%** compared with the same month in 2021.

Table 13. Macau - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2021 – LPG Rate (MOP/kg)	Year 2022 – LPG Rate (MOP/ kg)	% Change – LPG Rate
January	7.81	10.95	+40.3%
February	8.46	9.89	+16.9%
March	8.46	9.89	+16.9%
April	8.40	10.02	+19.3%
May	8.12	12.00	+47.8%
June	8.12	12.00	+47.8%
July	8.12	12.00	+47.8%
August			
September			
October			
November			
December			
Total/Average	**8.21**	**10.96**	**+33.5%**

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in July 2022 decreased by **32.3%** compared to 2019 normalized consumption in the same month.

Table 14. Macau - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2022 Actual LPG Consumption (kg)	% Change – Normalized
January	95,134	121,221	+27.4%
February	117,728	146,623	+24.5%
March	78,544	91,040	+15.9%
April	85,634	85,743	+0.1%
May	70,235	82,704	+17.8%
June	72,880	77,044	+5.7%
July	66,446	44,991	-32.3%
August			
September			
October			
November			
December			
Total/Average	**586,602**	**649,365**	**+10.7%**

2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to July 2022 is **5.9%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

Table 15. Macau ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to July 2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-5.9%**	**-5.9%**	-	-	-

Table 16. Macau ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2022 Actual Energy Consumption (GJ)	% Change – Normalized vs Actual
January	21,555	22,715	+5.4%
February	22,007	22,000	-0.03%
March	22,792	22,672	-0.5%
April	24,151	22,358	-7.4%
May	24,532	23,470	-4.3%
June	26,589	24,287	-8.7%
July	27,705	21,769	-21.4%
August			
September			
October			
November			
December			
Total/Average	**169,332**	**159,273**	**-5.9%**

Important note: *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 17, the actual water consumption and cost in July 2022 decreased by **32.6%** and **32.5%**, respectively compared to the same month in 2021. To date, MGM Macau has decreased actual water consumption and associated costs by **10.5%** and **10.3%** respectively.

Table 17. Macau - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2021	2022	% Change Actual	2021	2022	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	39,952	37,078	-7.2%	299,330	287,500	-4.0%	+15.8%	+36.7%
February	41,079	41,524	+1.1%	317,780	322,270	+1.4%	-90.8%	+43.2%
March	43,168	39,962	-7.4%	334,000	310,160	-7.1%	-6.9%	-9.6%
April	49,819	45,295	-9.1%	386,600	351,540	-9.1%	-11.4%	-11.4%
May	53,145	49,042	-7.7%	420,070	380,590	-9.4%	-32.6%	-12.2%
June	44,508	40,383	-9.3%	345,320	313,390	-9.2%	-1.0%	-35.0%
July	46,227	31,179	-32.6%	358,640	242,100	-32.5%	+4.1%	-92.5%
August								
September								
October								
November								
December								
Total/Average	**317,898**	**284,463**	**-10.5%**	**2,461,740**	**2,207,550**	**-10.3%**	**-13.1%**	**-15.5%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Podium water consumption for July 2022 decreased by 27.7% compared to the same month in 2021.
- Hotel water consumption for July 2022 decreased by 59.9% compared to the same month in 2021. Moreover, the guest occupancy decreased by 89.4% compared to the same month in 2021.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Table 18 presents the comparison of water rates in the years 2021 and 2022. The water rate in July 2022 increased by **0.09%** compared to the same month in 2021. The average utility rate of water has increased by **0.3%** compared to 2021.

Table 18. Macau - Summary of Actual Monthly Water Rates

Monthly Period	Year 2021 – Water Rate (MOP/m³)	Year 2022 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.49	7.75	+3.5%
February	7.74	7.76	+0.3%
March	7.74	7.76	+0.3%
April	7.760	7.761	+0.01%
May	7.90	7.76	-1.8%
June	7.759	7.76	+0.02%
July	7.758	7.765	+0.09%
August			
September			
October			
November			
December			
Total/Average	**7.74**	**7.76**	**+0.3%**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in July 2022 decreased by **14.9%** compared to 2019 normalized consumption in the same month.

Table 19. Macau - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2022 Actual Water Consumption (m³)	% Change - Normalized
January	35,691	37,078	+3.9%
February	37,518	41,524	+10.7%
March	37,955	39,962	+5.3%
April	39,506	45,295	+14.7%
May	42,691	49,042	+14.9%
June	41,208	40,383	-2.0%
July	36,632	31,179	-14.9%
August			
September			
October			
November			
December			
Total/Average	**271,201**	**284,463**	**+4.9%**

2.5 Macau Year 2022 Projects

Energenz will continue to perform analysis and coordinate with MGM to implement the following projects in 2022 and reduce the site energy and water consumption which supports Energenz annual normalized energy savings target.

Table 20. MGM Macau Year 2022 Projects

Item	Initiatives
Electricity & LPG	
1	Chilled Water Pump Retrofit (Phase 1)
2	AHU Static Pressure Set-Point Reset (Phase 2)
3	PICCV Implementation (Phase 3)
4	Air Side Equipment EC Fan/Motor Replacement
5	VAV Space Temperature Setpoint Optimization with Heatmap Control
6	Heatmap Phasing Implementation (Stage 3, 9a, 9b, 10a)
7	Neon Light LED Replacement (Phase 2)
8	Replacement of L2M Steam-to-Water Heat Exchanger to Fully Utilize Heat Pump
Water	
9	CT Conductivity Blowdown Optimization

3. MGM Cotai Performance

3.1 Electricity

The actual electricity consumption in July 2022 decreased by **23.4%** and the cost decreased by **20.1%** compared to the same month in 2021. To date, MGM Cotai has decreased actual electricity consumption by **9.4%** and associated costs decreased by **5.3%**.

Table 21. Cotai - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	8,168,400	8,336,400	+2.1%	9,315,910	10,013,460	+7.5%	+15.8%
February	7,637,640	7,528,560	-1.4%	8,737,740	9,066,060	+3.8%	-90.8%
March	9,197,160	8,833,200	-4.0%	10,499,490	10,632,040	+1.3%	-6.9%
April	9,517,560	8,985,360	-5.6%	11,430,660	11,262,370	-1.5%	-11.4%
May	11,446,440	9,585,000	-16.3%	13,735,880	11,983,070	-12.8%	-32.6%
June	11,122,320	10,087,080	-9.3%	13,340,650	12,615,080	-5.4%	-1.0%
July	11,853,120	9,082,680	-23.4%	14,194,900	11,346,140	-20.1%	+4.1%
August							
September							
October							
November							
December							
Total/Average	**68,942,640**	**62,438,280**	**-9.4%**	**81,255,230**	**76,918,220**	**-5.3%**	**-13.1%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Due to the adjusted operations during COVID low occupancy, the overall site electricity has decreased significantly when compared with the same month last year.

Table 22 presents the monthly e-bus electricity consumption in MGM Cotai. There was no e-bus electricity consumption in July 2022.

Table 22. Cotai - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2021	2022	% Change - Actual
January	44,839	42,287	-5.7%
February	32,534	47,905	+47.2%
March	34,902	38,793	+11.1%
April	37,412	38,537	+3.0%
May	51,584	51,328	-0.5%
June	56,629	36,420	-35.7%
July	64,410	0	-100.0%
August			
September			
October			
November			
December			
Total/Average	**322,309**	**255,269**	**-20.8%**

Table 23 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in July 2022 increased by **4.3%** compared to the same month in 2021.

Table 23. Cotai - Summary of Actual Monthly Electricity Rates

Monthly Period	Year 2021 – Electricity Rate (MOP/kWh)	Year 2022 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.14	1.20	+5.3%
February	1.14	1.20	+5.3%
March	1.14	1.20	+5.4%
April	1.20	1.25	+4.4%
May	1.20	1.25	+4.2%
June	1.20	1.25	+4.3%
July	1.20	1.25	+4.3%
August			
September			
October			
November			
December			
Total/Average	**1.17**	**1.23**	**+4.7%**

Table 24 summarized the monthly normalized electricity consumption. The actual electricity consumption in July 2022 decreased by **22.9%** compared to 2019 normalized consumption in the same month.

Table 24. Cotai - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2022 Actual Electricity Consumption (kWh)	% Change - Normalized
January	8,635,397	8,336,400	-3.5%
February	8,394,419	7,528,560	-10.3%
March	9,473,790	8,833,200	-6.8%
April	9,911,574	8,985,360	-9.3%
May	10,366,415	9,585,000	-7.5%
June	11,183,378	10,087,080	-9.8%
July	11,776,857	9,082,680	-22.9%
August			
September			
October			
November			
December			
Total/Average	**69,741,829**	**62,438,280**	**-10.5%**

3.2 Natural Gas Consumption

The actual natural gas consumption and cost in July 2022 decreased by **97.0%** and **96.8%** respectively compared to the same month in 2021. To date, MGM Cotai has decreased actual natural gas consumption and associated costs by **42.1%** and **38.7%**, respectively.

Table 25. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual Natural Gas Consumption (m^3)			Actual Natural Gas Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost		
January	60,643	59,378	-2.1%	363,990	376,240	+3.4%	+56.0%	+22.3%
February	54,221	76,089	+40.3%	325,930	482,130	+47.9%	-52.6%	+31.6%
March	68,066	52,146	-23.4%	407,980	330,410	-19.0%	-5.4%	-34.3%
April	86,866	43,900	-49.5%	518,960	278,170	-46.4%	-28.6%	-37.5%
May	90,102	45,632	-49.4%	537,940	289,140	-46.3%	-26.8%	-48.1%
June	80,606	33,949	-57.9%	482,270	215,120	-55.4%	-1.2%	-55.1%
July	102,473	3,041	-97.0%	610,450	19,270	-96.8%	-0.8%	-92.5%
August								
September								
October								
November								
December								
Total/Average	**542,977**	**314,135**	**-42.1%**	**3,247,520**	**1,990,480**	**-38.7%**	**-12.8%**	**TBC%**

Important note: *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption decrease below:

- Kitchen NG consumption has decreased by 93.9% and food cover decreased by 94.2% compared to the same month in 2021.
- NG boiler consumption has decreased by 99.96% compared to the same month in 2021.

Table 26 presents the comparison of natural gas rates in the years 2021 and 2022. The natural gas rate in July 2022 increased by **6.4%** compared with the same month in 2021.

Table 26. Cotai - Summary of Actual Monthly Natural Gas Rates

Monthly Period	Year 2021 – Natural Gas Rate (MOP/m³)	Year 2022 – Natural Gas Rate (MOP/m³)	% Change – Natural Gas Rate
January	6.00	6.34	+5.6%
February	6.01	6.34	+5.4%
March	5.99	6.34	+5.7%
April	5.97	6.34	+6.1%
May	5.97	6.34	+6.1%
June	5.98	6.34	+5.9%
July	5.96	6.34	+6.4%
August			
September			
October			
November			
December			
Total/Average	**5.98**	**6.34**	**+5.9%**

Table 27 summarized the monthly normalized natural gas consumption. The actual natural gas consumption in July 2022 decreased by **97.1%** compared to 2019 normalized consumption in the same month. The normalized model and comparison of natural gas has been modified and included since January 2022.

Table 27. Cotai - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2022 Actual Natural Gas Consumption (m³)	% Change - Normalized
January	125,664	59,378	-52.7%
February	132,723	76,089	-42.7%
March	119,389	52,146	-56.3%
April	122,527	43,900	-64.2%
May	116,565	45,632	-60.9%
June	105,271	33,949	-67.8%
July	103,702	3,041	-97.1%
August			
September			
October			
November			
December			
Total/Average	**825,841**	**314,135**	**-62.0%**

3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included in the energy consumption tracking as part of the ISO50001 requirement. To date, normalized energy saving achieved up to July 2022 is **10.0%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. Cotai ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to July 2022	2023	2024	2025
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.8%**	**-10.0%**	-	-	**-**

Table 29. Cotai ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2022 Actual Energy Consumption (GJ)	% Change - Normalized vs Actual
January	35,448	32,071	-9.5%
February	34,825	29,743	-14.6%
March	38,248	33,609	-12.1%
April	39,933	33,871	-15.2%
May	41,364	36,089	-12.8%
June	43,913	37,492	-14.6%
July	45,995	32,803	-28.7%
August			
September			
October			
November			
December			
Total/Average	**279,727**	**235,678**	**-15.7%**

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

3.4 Water Consumption

According to Table 30, the actual water consumption in July 2022 decreased by **46.2%** compared to the same month in 2021. To date, MGM Cotai has decreased actual water consumption and associated costs by **24.1%** and **24.3%** respectively.

Table 30. Cotai - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2021	2022	% Change Actual	2021	2022	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	43,292	44,658	+3.2%	336,700	347,240	+3.1%	+15.8%	+22.3%
February	50,837	45,298	-10.9%	395,180	352,330	-10.8%	-90.8%	+31.6%
March	55,668	44,914	-19.3%	432,400	349,260	-19.2%	-6.9%	-34.3%
April	65,511	51,640	-21.2%	509,000	401,500	-21.1%	-11.4%	-37.5%
May	77,506	55,628	-28.2%	612,200	432,480	-29.4%	-32.6%	-48.1%
June	66,115	45,805	-30.7%	513,500	356,090	-30.7%	-1.0%	-55.1%
July	70,716	38,031	-46.2%	549,160	295,890	-46.1%	+4.1%	-95.5%
August								
September								
October								
November								
December								
Total/ Average	**429,645**	**325,974**	**-24.1%**	**3,348,140**	**2,534,790**	**-24.3%**	**-13.1%**	**-35.5%**

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption decreased by 80.3% due to 94.2% occupancy decrease when compared with the same month last year.
- CT make up water consumption decreased by 25.6% due to the decrease in the Chiller Plant operations when compared with the same month last year.

Table 31 presents the comparison of water rates in the years 2021 and 2022. The water rate in July 2022 increased by **0.2%** compared to the same month in 2021. The average utility rate of water has decreased by **0.2%** compared to 2021.

Table 31. Cotai - Summary of Actual Monthly Water Rates

Monthly Period	Year 2021 – Water Rate (MOP/m³)	Year 2022 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.777	7.776	-0.02%
February	7.77	7.78	+0.1%
March	7.77	7.78	+0.1%
April	7.77	7.775	+0.1%
May	7.90	7.77	-1.6%
June	7.767	7.774	+0.1%
July	7.77	7.78	+0.2%
August			
September			
October			
November			
December			
Total/Average	**7.79**	**7.78**	**-0.2%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in July 2022 decreased by **30.6%** compared to 2019 normalized consumption in the same month.

Table 32. Cotai - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2022 Actual Water Consumption (m³)	% Change - Normalized
January	48,361	44,658	-7.7%
February	49,015	45,298	-7.6%
March	49,249	44,914	-8.8%
April	51,413	51,640	+0.4%
May	54,331	55,628	+2.4%
June	56,037	45,805	-18.3%
July	54,785	38,031	-30.6%
August			
September			
October			
November			
December			
Total/Average	**363,190**	**325,974**	**-10.2%**

3.5 Cotai Year 2022 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2022 and reduce the site energy and water consumption which supports Energenz annual normalized energy savings target.

Table 33. MGM Cotai Year 2022 Projects

Item	Initiatives
Electricity & NG	
1	Chiller Plant Control Optimization (CHWP/CDWP/CT)
2	Hot Water System Upgrade 1. Increase Max. No of Plate for Pre-heat Compact Unit 2. Upgrade Booster Compact Unit 3. Install Electric Heater for Booster Calorifier
3	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)
4	PICCV Implementation (Phase 3)
5	Hotel Towers Building Pressurization and EAF Optimization
6	Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization
7	Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2)
8	VAV Space Temperature Setpoint Optimization with Heatmap Control
9	Heatmap Phasing Implementation (Stage 5, 6, 7, 8a, 8b, 10b)
10	Centralized RO Pump Set Replacement (10 Units)
Water	
11	Extension of CDP Water Recycling
12	Rainwater Harvesting for CT Make Up (Phase 1)
13	CT Conductivity Blowdown Optimization

Appendix A

Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2022.

MGM Macau

Utilities	2022	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM Cotai

Utilities	2022	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

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